AIR BAG SYSTEM [TWO-STEP DEPLOYMENT CONTROL SYSTEM]

id0810b1500100

Outline

- With the air bag system, the SAS control module operates (deploys) the air bags according to the degree of impact during the collision to supplement the occupant protection function of the seat belts.

Warning

- **The air bag system supplements the occupant protection function of the seat belts, however the intended effect of the air bag system cannot be obtained if the seat belts are not fastened correctly. When driving the vehicle, always fasten the seat belts correctly.**

Function

- With the air bag system, the SAS control module calculates the degree of impact based on the collision detection signal from each of the following sensors, and operates (deploys) the required parts.

×: Collision detection
××: Operation (deployment)
—: No detection

Collision area	Sensor				Operation (deployment) part					
	Crash sensor integrated into SAS control module	Crash sensor integrated into crash zone sensor	Crash sensor integrated into side air bag sensor	Pressure detection sensor integrated into pressure sensor	Driver-side air bag module	Passenger-side air bag module	Curtain air bag module (driver-side/ passenger-side)	Side air bag module (driver-side/ passenger-side)	Front pre-tensioner seat belt (driver-side/ passenger-side)	Rear pre-tensioner seat belt (driver-side/ passenger-side)
Collision from front of vehicle or frontal offset	×	—	—	—	—	—	—	—	—	—
	—	×	—	—	—	—	—	—	—	—
	×	×	—	—	××	××	××*1	××*1	××	××
Collision from side of vehicle	×	—	—	—	—	—	—	—	—	—
	—	—	×	—	—	—	—	—	—	—
	—	—	—	×	—	—	—	—	—	—
	×	—	×	—	—	—	××*2	××*2	××*2	—
	×	—	—	×	—	—	××*2	××*2	××*2	—
	×	—	×	×	—	—	××*2	××*2	××*2	—
Collision from rear of vehicle	×	—	—	—	—	—	—	—	—	—

*1 : Operates (deploys) by degree of impact during collision.

*2 : Operates (deploys) on the side the collision is detected.

Driver-side air bag deployment force switching function (two-step deployment)

- The driver's air bag deployment force switching function switches the driver's air bag deployment force by deploying inflators No.1 and 2 built into the driver's air bag module simultaneously or only inflator No.1*1 according to the degree of impact from a frontal or frontal offset collision, the slide position of the driver's seat, and the front seat belt fastening status.

*1 : If the SAS control module deploys only inflator No.1, it deploys inflator No.2 soon afterwards to not leave it undeployed.

Collision impact	Front and back positions of driver's seat	Front seat belt status	Driver-side air bag module		Air bag deployment force
			Inflator No.1	Inflator No.2	
Small	Position other than foremost position	Equipped	Deployment	—	Small
Large			Deployment	Deployment	Large
Small		Not equipped	Deployment	Deployment	Large
Large			Deployment	Deployment	Large

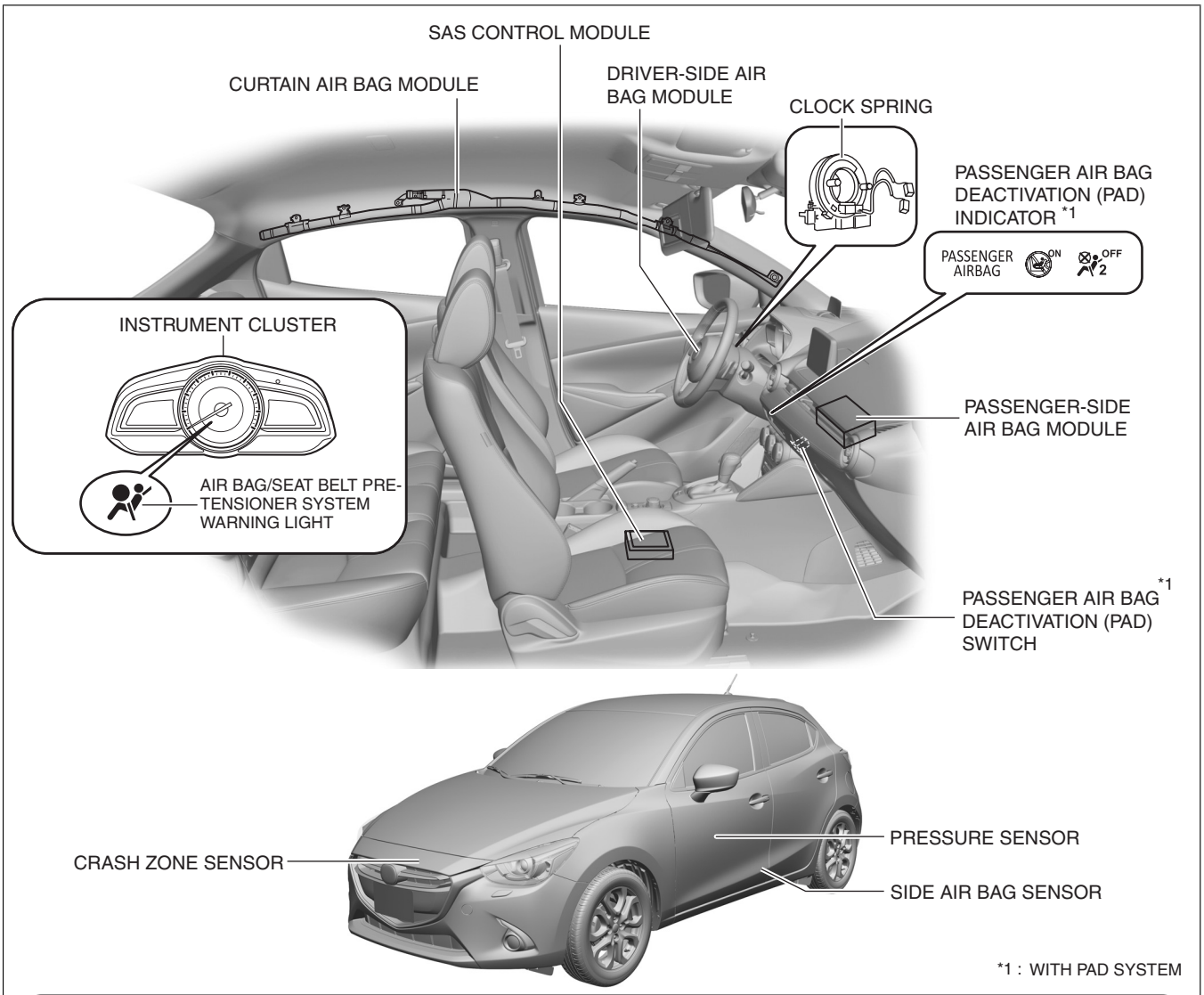
Collision impact	Front and back positions of driver's seat	Front seat belt status	Driver-side air bag module		Air bag deployment force
			Inflator No.1	Inflator No.2	
Small	Foremost position	Equipped	Deployment	—	Small
Large			Deployment	—	Small
Small		Not equipped	Deployment	—	Small
Large			Deployment	—	Small

System conditions display function

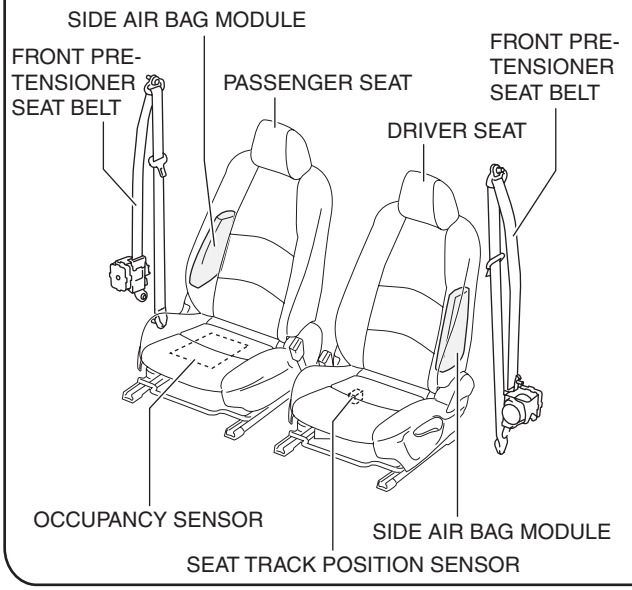
- The air bag system detects each part condition and the SAS control module sends the system status signal to the instrument cluster.
- The instrument cluster notifies the user of the system conditions using the following warning light or alarm based on the system condition signal.

Warning light and alarm	Reference
Air bag/front seat belt pre-tensioner system warning light	(See AIR BAG/SEAT BELT PRE-TENSIONER SYSTEM WARNING LIGHT [TWO-STEP DEPLOYMENT CONTROL SYSTEM].)
Air bag system warning alarm	(See AIR BAG SYSTEM WARNING ALARM [TWO-STEP DEPLOYMENT CONTROL SYSTEM].)
Seat belt warning light	(See SEAT BELT WARNING ALARM.) (See INDICATOR UNIT.)

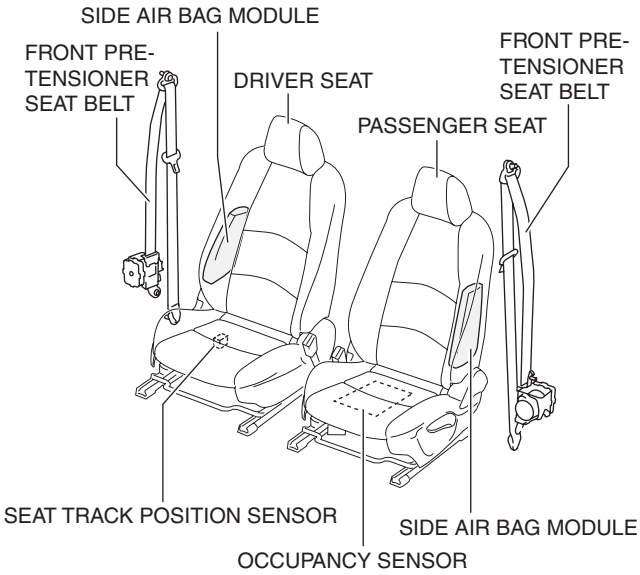
Structural View



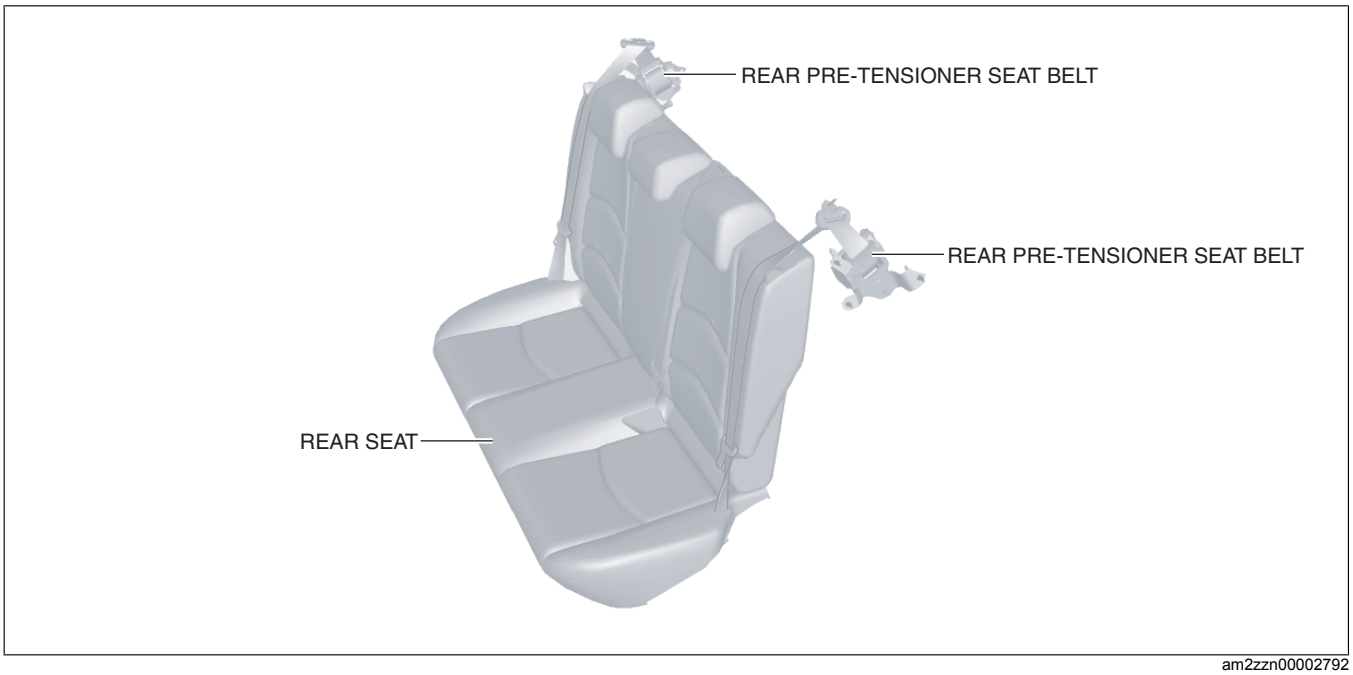
L.H.D.



R.H.D.

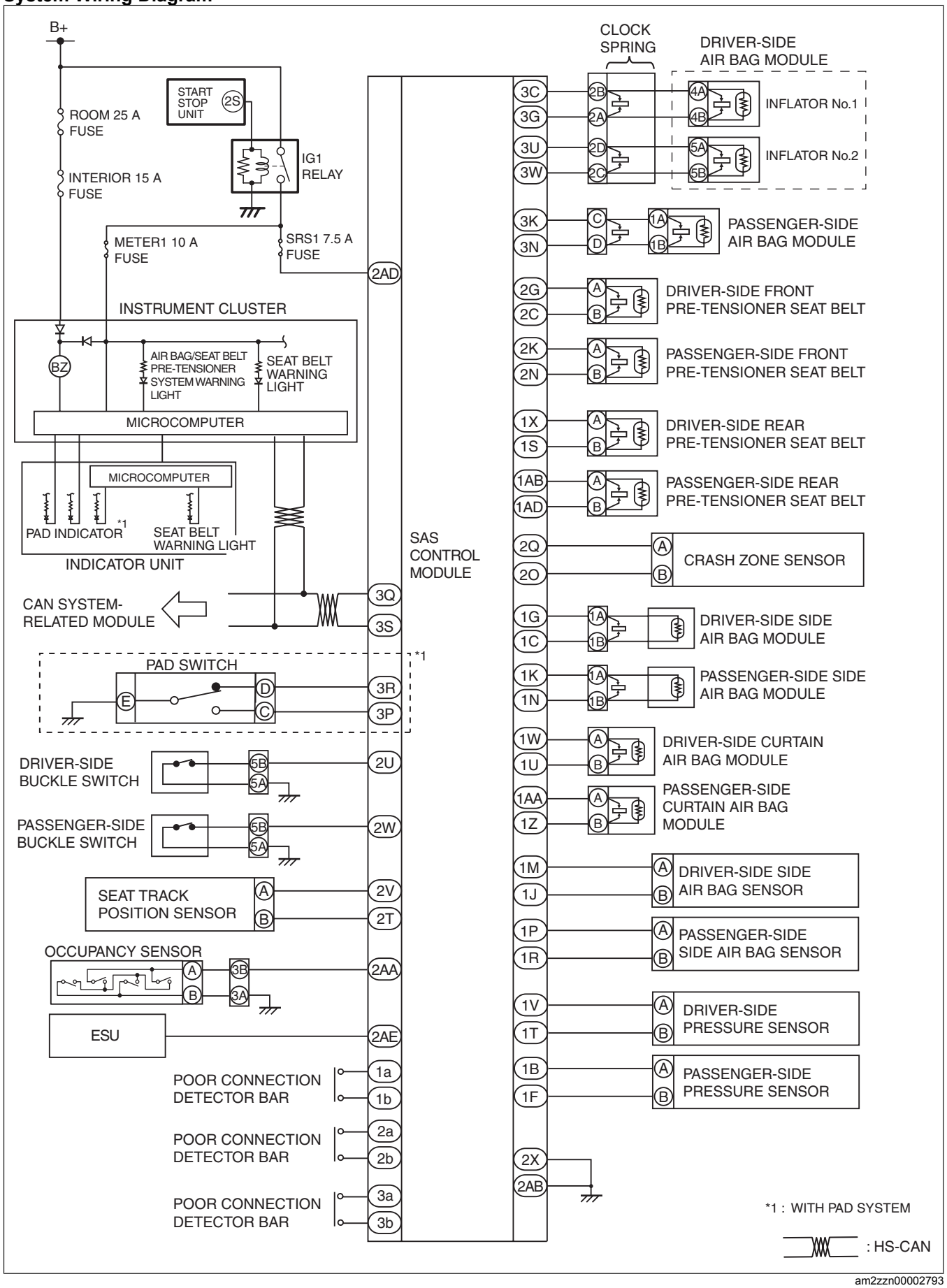


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System Wiring Diagram

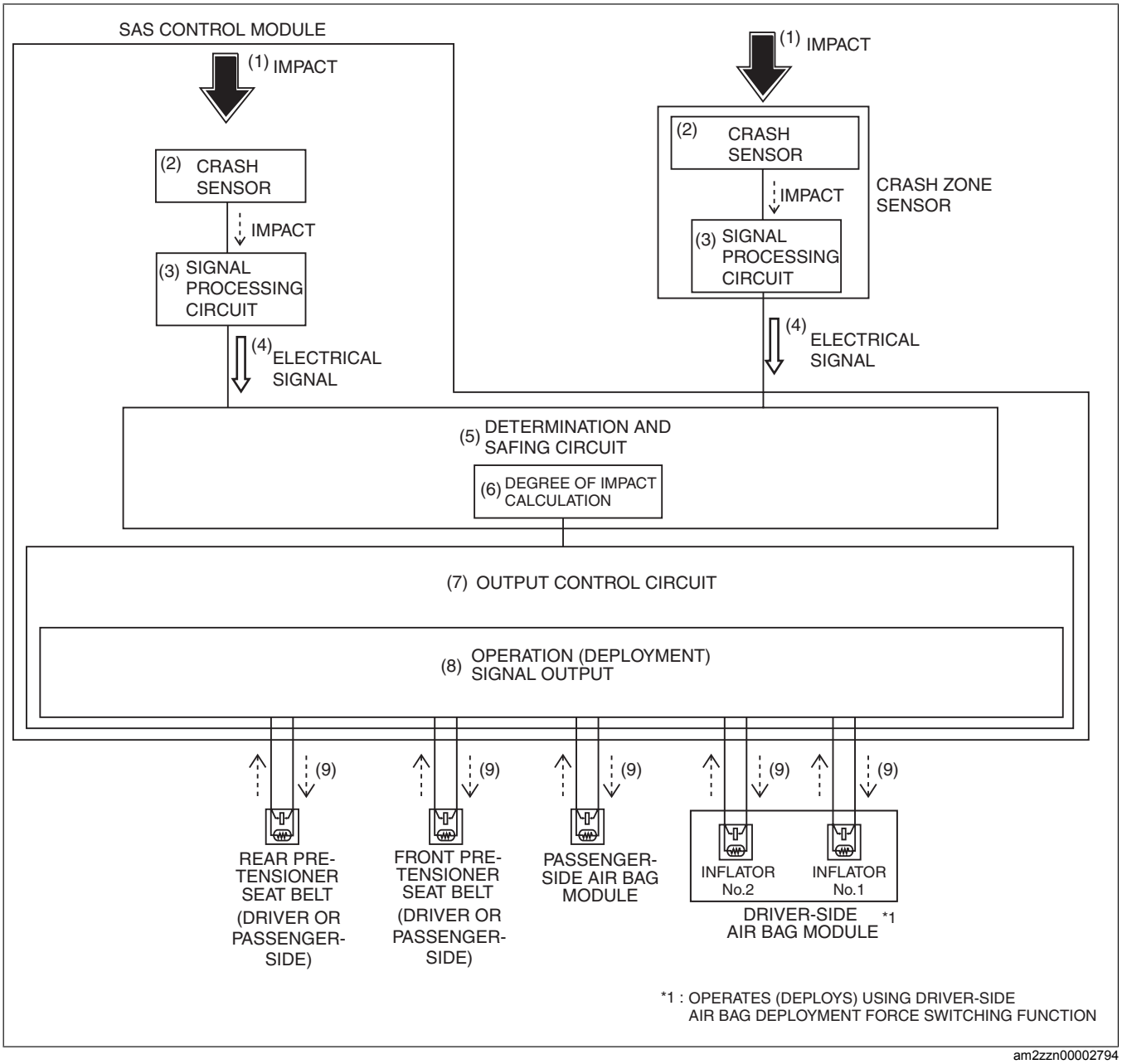


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Operation

Operation during frontal or frontal offset collision

1. When an impact from a frontal or frontal offset collision (1) is received, the crash sensor built into the crash zone sensor and the crash sensor (2) built into the SAS control module detect the impact and convert the impact into an electrical signal at the signal processing circuit (3).
2. The SAS control module calculates (6) the degree of impact using the determination and safing circuit (5) based on the electric signal (4) from each crash sensor.
3. If the calculated degree of impact exceeds the previously set threshold, the SAS control module outputs (8) an operation (deployment) signal to each air bag module or pre-tensioner seat belt at the output control circuit (7).
4. Each air bag module or pre-tensioner seat belt operates (deploys) (9) based on the operation (deployment) signal from the SAS control module.

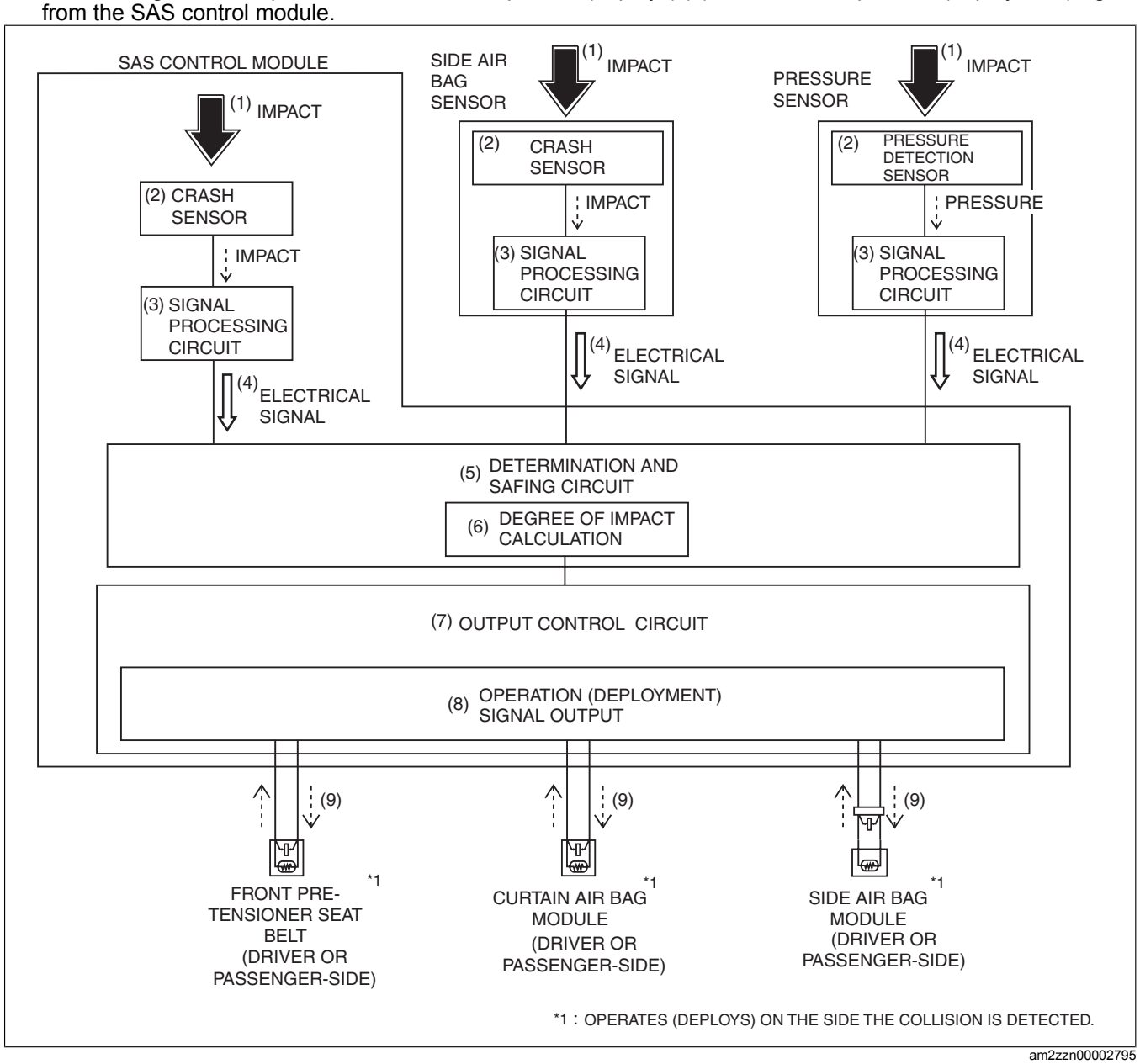


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Operation during a lateral collision

1. When an impact (1) from a lateral collision is received, the following sensors (2) detect the impact and convert the impact or the pressure into an electrical signal at the signal processing circuit (3).
   - — Crash sensor integrated into side air bag sensor
   - — Crash sensor integrated into SAS control module
   - — Pressure detection sensor integrated into pressure sensor
2. The SAS control module calculates (6) the degree of impact using the determination and safing circuit (5) based on the electric signal (4) from each crash sensor or the pressure detection sensor.

3. If the calculated degree of impact exceeds the previously set threshold, the SAS control module outputs (8) an operation (deployment) signal to each air bag module or pre-tensioner seat belt at the output control circuit (7).
4. Each air bag module or pre-tensioner seat belt operates (deploys) (9) based on the operation (deployment) signal from the SAS control module.



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